NORTHCOTE HIGH SCHOOL • YEAR 7/8 SCIENCE • VICTORIAN CURRICULUM 2.0

Space

Student Workbook

Ten sections — one for each lesson:

1 Earth, Moon & Sun • **2** Day, Night & Seasons
3 The Moon — Phases & Tides • **4** Reading the Rocks
5 Our Solar System • **6** Space Technology & Sustainability
7 Aboriginal & Torres Strait Islander Astronomy

Section 4 is optional — your teacher will tell you if your class is doing it.

Above level — extension
8 The Search for Life • **9** Gravity & Orbits • **10** Stars, Galaxies & Starlight

Name: ______________________ Class: __________

We acknowledge the Wurundjeri-Woiwurrung people of the Kulin Nation as the Traditional Custodians of the Country on which we learn.

Earth, Moon and Sun — Scale and Structure

VC 2.0: Science Understanding — Physical sciences: Earth and space sciences; scale and relative size of Earth, Moon and Sun

1.1 The scale of Earth, Moon and Sun

The three most important objects in our immediate cosmic environment differ enormously in size. Understanding their relative sizes is foundational to everything else in this unit.

Object	Diameter (km)	Mass (kg)	Distance from Earth
Moon	3,474	7.34×10^{22}	384,400 km avg
Earth	12,742	5.97×10^{24}	—
Sun	1,392,000	1.99×10^{30}	150,000,000 km (1 AU)

Key concept: Scale

If Earth were the size of a tennis ball (6.7 cm across), the Moon would be a marble (1.8 cm) about 1.9 m away, and the Sun would be a sphere 7.5 m across, placed 807 m down the road. The solar system is almost entirely **empty space**.

Check your understanding *8 marks*

Q1.1 (2 marks) Use the table to state the diameter of Earth and the diameter of the Sun.

Q1.2 (2 marks) Calculate how many times wider the Sun is than Earth. Show your working and give your answer to the nearest whole number.

Q1.3 (2 marks) The Moon is 384,400 km from Earth. Calculate how many Earth diameters would fit into that gap.

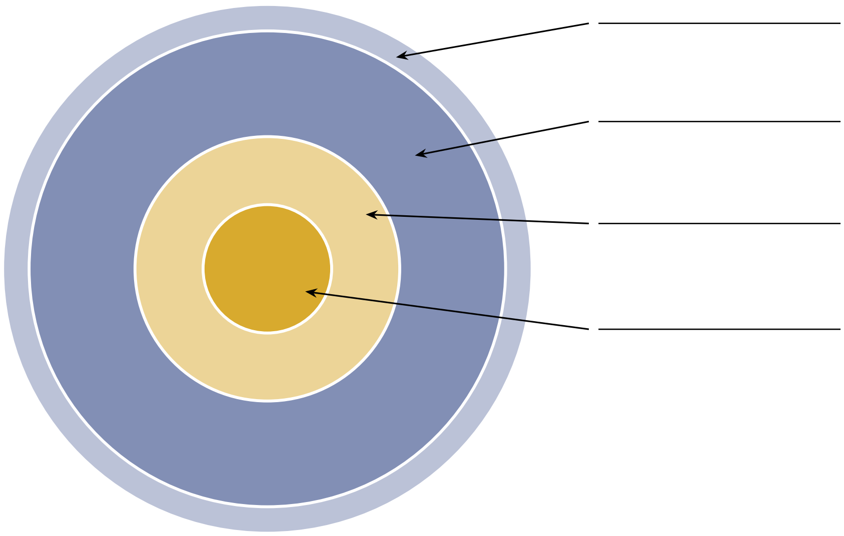
Q1.4 (2 marks) In the tennis-ball model, Earth is 6.7 cm across and the Sun sits 807 m down the road. Explain what this model shows that the table of diameters alone does not.

1.2 Structure of Earth

Earth has four main layers:

- **Inner core** — solid iron-nickel; radius 1,200 km; temperature 5,400°C
- **Outer core** — liquid iron-nickel; generates Earth's magnetic field (the dynamo effect)
- **Mantle** — semi-solid rock; 2,900 km thick; convection currents drive plate tectonics
- **Crust** — thin, solid rock; 5–70 km thick; oceanic (denser, thinner) and continental (less dense, thicker)

Diagram: Label the layers of Earth



Word bank: Outer core • Crust • Inner core • Mantle

Check your understanding *8 marks*

Q1.5 (2 marks) Which layer generates Earth's magnetic field, and what is this process called?

Q1.6 (2 marks) The mantle is described as semi-solid rock. State its thickness and what its convection currents drive.

Q1.7 (2 marks) Compare oceanic and continental crust using two differences given in the booklet.

1.3 The Moon

The Moon is Earth's only natural **satellite**. It produces no light of its own — everything we see is reflected sunlight.

- **Diameter**: 3,474 km — about one quarter of Earth's
- **Distance**: 384,400 km from Earth on average; light crosses this in **1.28 seconds**
- **Orbital period**: 27.3 days — and it *rotates* in exactly the same time
- **Surface**: covered in **regolith**, a fine powdery rock produced by billions of years of meteorite impacts
- **No atmosphere** and no liquid water, so temperatures swing from -173 °C at night to $+127$ °C in daylight

Why we only ever see one side

The Moon's rotation period and orbital period are identical (27.3 days), so the same face is always turned towards Earth. This is called **tidal locking**, and it is the normal outcome for a moon orbiting close to a much larger body. The hemisphere we never see from Earth is the **far side** — not the "dark side." It receives just as much sunlight as the near side.

Two kinds of terrain are visible with the naked eye. The number of craters on a surface tells you how old it is: the longer a surface has been there, the more impacts it has collected.

- **Highlands** — the bright, rugged regions. This is the Moon's **original crust**, so it is the oldest surface. It has been exposed to meteorite impacts for the whole history of the Moon, which is why it is so heavily cratered
- **Maria** (singular *mare*, Latin for "sea") — the dark, flat plains. Later, huge impacts punched basins into the crust, and lava welled up and flooded them. When that lava cooled it created a **new, smooth surface**, wiping out the craters that were there before. Because this happened after the highlands formed, the maria have had less time to collect impacts — so they are only lightly cratered

Check your understanding *9 marks*

Q1.8 (3 marks) Explain why we always see the same face of the Moon from Earth.

Q1.9 (2 marks) Describe the difference between the maria and the highlands, and state which is older.

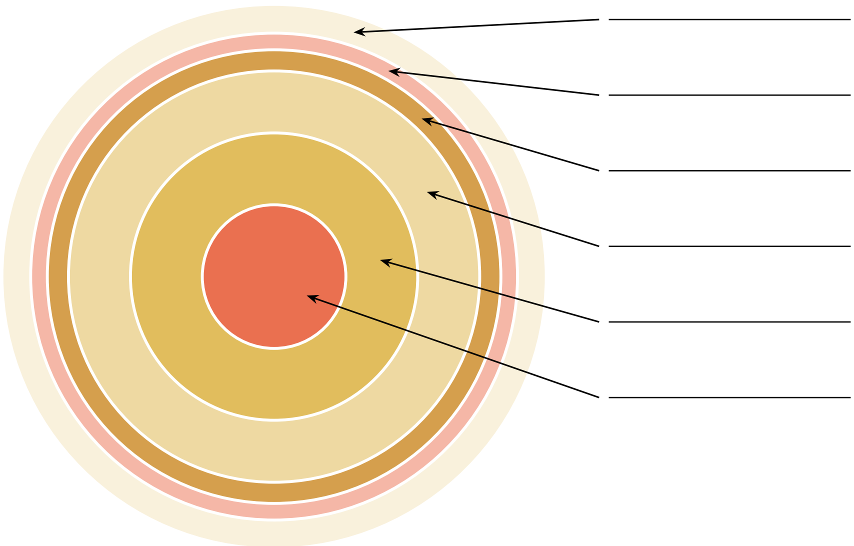
Q1.10 (2 marks) The Moon has no atmosphere. Explain how this causes its extreme temperature range.

1.4 Structure of the Sun

The Sun is a **main sequence star** — it converts hydrogen into helium through **nuclear fusion** in its core.

- **Core**: where fusion occurs; temperature 15 million°C
- **Radiative zone**: energy moves outward as photons (takes 100,000 years to cross)
- **Convective zone**: energy carried by convecting plasma
- **Photosphere**: the visible surface (5,778 K); where sunspots form
- **Chromosphere and Corona**: outer atmosphere; much hotter than photosphere (corona: 1–3 million K)

Diagram: Label the layers of the Sun



Word bank: Convective zone • Corona • Core • Photosphere • Chromosphere • Radiative zone

Worked example: Light travel time from the Sun

The Sun is 150,000,000 km from Earth. Light travels at 300,000 km/s. How long does sunlight take to reach us?

$$\text{Time} = \frac{\text{Distance}}{\text{Speed}} = \frac{150{,}000{,}000 \text{ km}}{300{,}000 \text{ km/s}} = 500 \text{ s} = 8 \text{ min } 20 \text{ s}$$

Answer: Sunlight takes 8 minutes and 20 seconds to reach Earth. This means the Sun we see right now is actually how it looked 8 minutes ago.

Your turn

A flash of light leaves the Moon. How long before we see it? (Moon is 384,400 km away; speed of light = 300,000 km/s)

Check your understanding *9 marks*

Q1.11 (2 marks) Define nuclear fusion as it occurs in the Sun.

Q1.12 (2 marks) State the temperature of the Sun's core and of its photosphere.

Q1.13 (3 marks) Jupiter is 5.20 AU from the Sun. Sunlight reaches Earth (1 AU) in 500 seconds. Calculate how long sunlight takes to reach Jupiter, in minutes and seconds.

Year 8 Extension: How the Moon formed — the Giant Impact Hypothesis

The leading model for the Moon's origin is the **Giant Impact Hypothesis**. About **4.5 billion years ago** a Mars-sized body — named **Theia** — struck the young Earth. The collision threw a disc of molten rock and vapour into orbit, and that material clumped together under gravity to form the Moon.

Two pieces of evidence support it:

- The Moon has almost **no iron core**. Earth's iron had already sunk to its centre before the impact, so the debris blasted into orbit came mostly from the rocky mantle.
- Moon rock returned by the Apollo missions has **oxygen isotope ratios identical to Earth's**. Rock formed elsewhere in the solar system does not match — so the Moon and Earth are made of the same material.

Why it matters: the Moon's mass stabilises Earth's axial tilt at about 23.5°. Without it, the tilt could wander between 0° and 85° over millions of years, making stable long-term climates — and possibly complex life — far less likely.

Year 8 Extension: Earth's magnetosphere and space weather

The liquid iron in Earth's **outer core** is constantly moving, and moving metal generates a magnetic field — the **dynamo effect**. That field does not stop at the surface. It reaches tens of thousands of kilometres out into space, forming the **magnetosphere**: an invisible shield around the whole planet.

It has something to shield us from.

- **Solar wind** — a constant stream of charged particles blowing outward from the Sun at 400–800 km/s. It never stops.
- **Coronal mass ejections** (CMEs) — huge bursts of plasma thrown off the Sun, which can reach Earth in one to three days.
- **Auroras** — the magnetosphere funnels charged particles toward the poles, where they collide with atmospheric gases and make them glow. This is the *aurora australis* in the south and the *aurora borealis* in the north. An aurora is what the shield looks like while it is working.

Space weather is the name for these events, and it has consequences on the ground. On 13 March 1989 a geomagnetic storm induced currents in Quebec's power transmission lines large enough to collapse the entire provincial grid in under 90 seconds, leaving about six million people without electricity for nine hours. Modern GPS, satellites and radio communication are all vulnerable.

Extension challenge. Mars has no global magnetic field, and most of its atmosphere has been stripped away by the solar wind over billions of years. Earth has both a field and an atmosphere. What does the comparison suggest the magnetic field is doing — and what might happen if Earth's field weakened a great deal?

Day, Night and Seasons

VC 2.0: Science Understanding — Earth and space sciences: Earth's rotation and revolution; axial tilt and seasons

2.1 Why we have day and night

Earth **rotates** (spins on its axis) once every **24 hours**. The side facing the Sun experiences day; the opposite side experiences night.

- Earth rotates **west to east** (anticlockwise when viewed from above the North Pole)
- This is why the Sun appears to rise in the east and set in the west
- Earth also **revolves** (orbits) around the Sun once every 365.25 days — one year

Draw it: day and night

Copy the simple diagram showing **Earth and the Sun**. On your diagram:

- shade the half of Earth where it is **night**, and label the half where it is **day**
- draw an arrow showing the direction of the Sun's light
- add a curved arrow showing which way Earth **rotates**

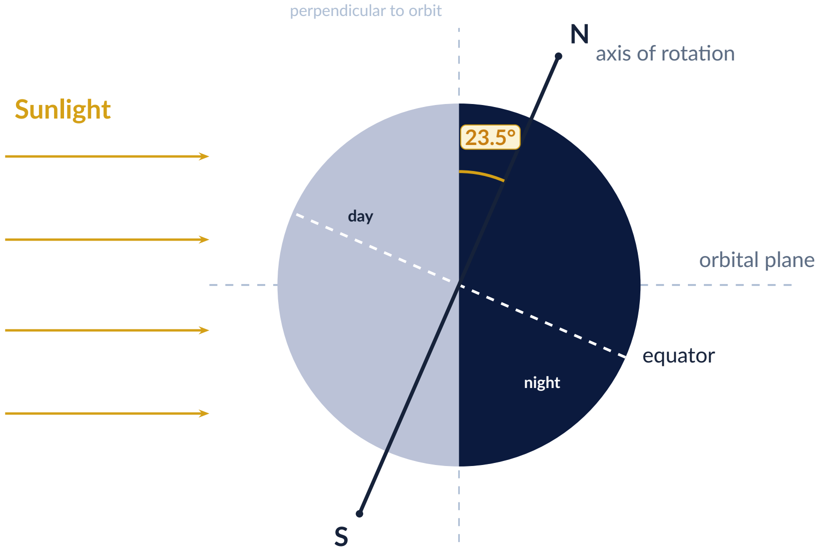
Check your understanding *4 marks*

Q2.1 (2 marks) State how long Earth takes to rotate once on its axis, and how long it takes to revolve once around the Sun.

Q2.2 (2 marks) Explain why the Sun appears to rise in the east and set in the west.

2.2 Why we have seasons

Seasons are caused by Earth's **axial tilt** of 23.5°, not by distance from the Sun. (Earth is actually closest to the Sun in January — Australian summer.)



The axis always points the same way, all year round.

Why tilt causes seasons

When the Southern Hemisphere tilts toward the Sun (December):

- The Sun is higher in the sky — its rays hit at a more direct angle (more energy per m^2)
- Days are longer — more hours of sunlight
- The combination produces **summer** in Australia

When the Southern Hemisphere tilts away (June): rays hit at a shallower angle, days are shorter — **winter**.

Check your understanding

6 marks

Q2.3 (1 mark) State the size of Earth's axial tilt.

Q2.4 (2 marks) State two effects of the Southern Hemisphere being tilted toward the Sun in December.

Q2.5 (3 marks) Earth is actually closest to the Sun in January. Explain why this is *not* the cause of Australian summer.

2.3 Solstices and equinoxes

Event	Date (approx.)	What happens
December solstice	21–22 December	Southern Hemisphere longest day; summer solstice for Australia
June solstice	21–22 June	Southern Hemisphere shortest day; winter solstice for Australia
March equinox	20–21 March	Equal day/night worldwide; autumn begins in Australia
September equinox	22–23 September	Equal day/night worldwide; spring begins in Australia

Your turn

Apply your knowledge — attempt all three, then we'll reveal answers:

Q1: It is March 21 (equinox) in Melbourne. Is it day or night at the North Pole right now?

Q2: A student in Brazil says "We're having summer right now." What month is it most likely to be?

Q3: Earth is actually slightly closer to the Sun in January than in July. Why is it still winter in Scandinavia in January?

Check your understanding *4 marks*

Q2.6 (2 marks) Using the table, state the date of the Australian summer solstice and what happens on that day.

Q2.7 (2 marks) State what is equal at an equinox, and name the Australian season that begins at the March equinox.

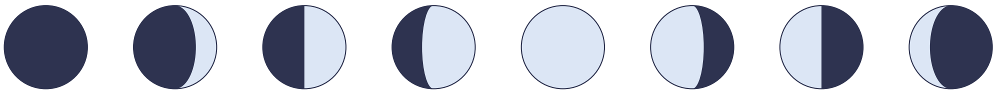
The Moon — Phases and Tides

VC 2.0: Science Understanding — Earth and space sciences: Moon phases and tidal patterns; gravitational effects

3.1 Why the Moon appears to change shape

The Moon does not produce light — it **reflects sunlight**. At all times, the half facing the Sun is lit. As the Moon orbits Earth over **29.5 days**, we see different fractions of the lit half from Earth — these are the **phases**.

The eight phases (in order)



← 29.5 days — one complete lunar cycle →

Common misconception

Earth's shadow does **NOT** cause phases. A **lunar eclipse** (Earth's shadow on the Moon) is a separate, less frequent event. Phases are caused by our changing view of the Moon's lit half.

Check your understanding *6 marks*

Q3.1 (1 mark) State how long the Moon takes to complete one full cycle of phases.

Q3.2 (3 marks) Explain why the Moon appears to change shape over a month.

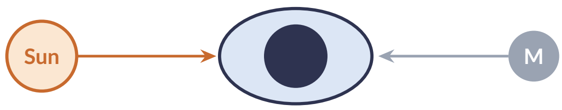
Q3.3 (2 marks) A student writes: "The phases of the Moon are caused by Earth's shadow falling on it." Correct this statement.

3.2 Tides

The Moon's gravity pulls on Earth's oceans. This creates a **tidal bulge** on the Moon-facing side, and a second bulge on the opposite side.

- **Spring tides**: Sun, Earth, and Moon are aligned (new or full Moon). Gravitational pulls add together — very large tidal range
- **Neap tides**: Moon at right angles to the Sun–Earth line (quarter phases). Pulls partially cancel — moderate tidal range
- "Spring" refers to the Old English word for "to leap," not the season

Spring tide (aligned)



Both pull same line → very large tidal range

Your turn: spring and neap tides

There is a **full Moon** on 5 March.

(a) What kind of tides would you expect that week — spring or neap? Explain your answer using the positions of the Sun, Earth and Moon.

(b) Roughly how many days later would you next expect **neap** tides, and what phase will the Moon be then?

(c) Why does one place on Earth get **two** high tides in a day, rather than just one?

Check your understanding *4 marks*

Q3.4 (1 mark) State what causes the tidal bulge on the side of Earth facing the Moon.

Q3.5 (3 marks) Explain the difference between spring tides and neap tides, including when each occurs.

Year 8 Extension: Eclipse geometry
Eclipses don't happen every New or Full Moon because the Moon's orbit is tilted 5.1° relative to Earth's orbit around the Sun. Eclipses only occur when a New or Full Moon coincides with the Moon being at a **node** (one of two points where its orbit crosses the ecliptic). The pattern of eclipses repeats every 18 years, 11 days, 8 hours — the **Saros cycle**.

Why the Moon fits the Sun perfectly: The Moon is 400× smaller than the Sun but 400× closer. Their angular diameters in our sky are almost identical — making total solar eclipses possible. This is unique in our solar system.

Reading the Rocks — Tides in Stone and the Acraman Impact

VC 2.0: Science as a Human Endeavour — Science knowledge is used to inform decisions and is developed through the work of individual scientists; Earth and space sciences: predictable phenomena

Everything so far in this unit has been observed by looking *up*. This section does the opposite. Two sets of South Australian rocks, both studied by **Professor George Williams** at the University of Adelaide, record events that happened in space — one a rhythm repeated for sixty years, the other a single morning that changed the planet. **We hold samples of both**, so this section is written to be worked through with the specimens in front of you.

4.1 Tides written in stone

Entry task: what could you count in a rock?

The photograph on the poster shows a slab of rock made of hundreds of thin, regular layers, each one a fraction of a millimetre thick. Before you read on: what natural process could lay down a layer of sediment that regularly, and what would counting the layers tell you?

A **tidal rhythmite** is a sedimentary rock made of finely layered silt and sand laid down by tidal currents. Each incoming tide carries sediment into a tidal inlet and drops it as a thin layer, called a **lamina**. A stronger tide carries more sediment and leaves a thicker lamina.

That makes a rhythmite a **recording** of the tides, in the same way a tree's rings record its years:

- one **lamina** = one tidal cycle (roughly one lunar day)
- laminae get thicker and thinner on a fortnightly cycle — the **spring** and **neap** tides you met in Section 2
- counting laminae between one neap and the next gives the number of days in a lunar month, *at the time the rock formed*

The **Elatina Formation** in the Flinders Ranges is a rhythmite **635 million years** old. Williams analysed a 10 m drill core holding a continuous 60-year record and counted:

Measurement	635 Ma	Today
Solar days per lunar month	30.5 ± 0.5	29.53
Lunar months per year	13.1 ± 0.1	12.37
Solar days per year	400 ± 7	365.24
Length of a solar day (hours)	21.9 ± 0.4	24.00

The year did not change — the day did

$400 \times 21.9 = 8760$ hours. $365.24 \times 24.00 = 8766$ hours. The two agree to better than 0.1%. Earth took just as long to go round the Sun 635 million years ago as it does now — it simply **spun faster**, so more days fitted into the same year.

Why Earth is slowing down

The Moon raises a tidal bulge in Earth's oceans. Earth rotates faster than the Moon orbits, so friction drags that bulge slightly *ahead* of the Moon. The bulge pulls the Moon forward, and the Moon pulls back on Earth.

- Earth loses rotational energy — **the day gets longer**
- the Moon gains orbital energy — **the Moon moves further away**

This is called **tidal braking**. Laser reflectors left on the Moon by Apollo astronauts let us measure the recession directly today: **3.8 cm per year**.

The rhythmites let us check that figure against the deep past, and the answer is interesting: averaged over the last 635 million years the Moon receded at only about **2 cm per year** — barely half today's rate. The rate is not a constant. It depends on the shape of the ocean basins, which changes as the continents drift.

Why this matters for where the Moon came from

Any explanation of the Moon's origin has to produce an orbit that behaves the way the rocks say it behaved. Assume today's recession rate has always applied, wind the clock back, and you get an Earth–Moon history that does not fit a 4.5-billion-year-old Moon. The rhythmites are one of very few **direct measurements** of the ancient Earth–Moon system, so they are used to test the models rather than trust them. The currently accepted explanation — a Mars-sized body striking the young Earth and throwing debris into orbit — rests mostly on other evidence (the Moon's tiny iron core, and lunar rock chemistry that matches Earth's mantle almost exactly).

Check your understanding

13 marks

Q4.1 (1 mark) State what one lamina in a tidal rhythmite represents.

Q4.2 (2 marks) Explain why laminae in a rhythmite vary in thickness on a fortnightly cycle.

Q4.3 (2 marks) Using the table, state how many days there were in a year 635 million years ago, and how long a day was.

Q4.4 (3 marks) Show that a 400-day year of 21.9-hour days is about the same length as our year of 365.24 days. Explain what this tells you about Earth's orbit compared with Earth's spin.

Q4.5 (3 marks) Describe how tidal braking makes Earth's day longer and pushes the Moon further away. Refer to the tidal bulge in your answer.

Q4.6 (2 marks) The Moon currently recedes at 3.8 cm per year, but the average over the last 635 million years was about 2 cm per year. State what this shows about the recession rate, and give one reason it changes.

4.2 The Acraman impact

Entry task: what would be left behind?

An asteroid 5 km across hits South Australia. 580 million years of erosion follow. List three things you would look for today as evidence that it happened.

About **580 million years ago** an asteroid roughly **5 km across**, travelling at **25 km per second**, struck the Gawler Range Volcanics in what is now the Gawler Ranges of South Australia.

Quantity	**Value**
Diameter of the asteroid	≈5 km
Impact speed	25 km/s
Energy released	5 million megatons of TNT
Transient crater	40 km across, 4 km deep
Final collapse crater	90 km across
Ejecta found	up to 300 km away

It is the **largest known impact structure in Australia**, and it disturbed the global environment.

Finding a crater that is not there

The crater itself has been erased by 580 million years of erosion. What survives is a circular topographic depression **30 km across**, underlain by shattered volcanic rock, with the salt lake **Lake Acraman** (20 km across) sitting inside it. Satellite infra-red imagery shows the wider structure as a set of concentric rings 90 km in diameter.

The impact was dated not from the crater but from the debris. Fragments of Gawler Ranges dacite — a volcanic rock found nowhere near the Flinders Ranges — turn up in Flinders Ranges shales **300 km away**, in a layer 580 million years old. That layer dates the impact, and the same fallout horizon has been identified overseas.

Why an impact leaves rock that looks like nothing else

At 25 km/s the asteroid delivers more energy than the rock can absorb as heat or motion, so the rock responds in ways it never does otherwise: it **shatters** along cone-shaped fracture surfaces, it **melts** instantly beneath the point of impact, and quartz crystals develop **planar deformation features** — microscopic parallel lines that form only under shock pressure. These are the fingerprints geologists use to identify an impact site anywhere on Earth, and on the Moon.

The specimens

Handle each sample, then complete the table. Look for grain size, angular fragments, colour, layering, and anything that looks as though it flowed.

Specimen	What I can see	What it tells me about the impact
Shattered dacite		
Melted dacite		
Ejecta in Flinders shale		
Disturbed sediment with dark fallout grains		

Check your understanding *13 marks*

Q4.7 (2 marks) State the diameter and impact speed of the Acraman asteroid.

Q4.8 (2 marks) Explain why the Acraman crater cannot be seen today, and state what can be seen instead.

Q4.9 (3 marks) Dacite from the Gawler Ranges is found in Flinders Ranges shale 300 km away. Explain how this both proves an impact occurred and fixes its date.

Q4.10 (2 marks) Name two features of a rock that show it has been through an impact rather than ordinary geological processes.

Q4.11 (2 marks) The transient crater was 40 km across but the final crater was 90 km across. Suggest why the crater ended up wider than the hole first excavated.

Q4.12 (2 marks) The Moon is covered in craters and Earth is not, even though both have been hit about as often. Give two reasons Earth's craters are missing.

Year 8 Extension: How much closer was the Moon?
Kepler's third law says the square of an orbital period is proportional to the cube of the orbital radius: $T^2 \propto r^3$, so $r \propto T^{2/3}$.
At 635 Ma there were 13.1 lunar months in a year; today there are 12.37. Since the year itself has not changed length, the month was $12.37/13.1 = 0.944$ times as long as it is now. Therefore

$$\frac{r_{635}}{r_{\text{now}}} = 0.944^{2/3} = 0.963$$

so the Moon sat at about $0.963 \times 384{,}400 \approx 370{,}000$ km — roughly **14,000 km closer**, or 3.7%. Spread over 635 million years that is an average recession of about 2.2 cm per year, which is where the figure in Section 3.1 comes from. Notice what this does *not* say: the Moon was not dramatically closer in the Precambrian. The interesting result is the **rate**, not the distance.

Try it: the Moon's distance is measured today as 384,400 km and it recedes at 3.8 cm per year. If that rate had always applied, how far away was the Moon 635 million years ago? Compare your answer with the 14,000 km above, and say what the difference tells you.

Source and acknowledgement

Both topics in this section come from the research of **Professor George E. Williams**, Department of Earth Sciences, University of Adelaide. The specimens and research posters were provided to Northcote High School by **Stuart Williams**.

Williams, G.E. (2000). Geological constraints on the Precambrian history of Earth's rotation and the Moon's orbit. *Reviews of Geophysics* 38, 37–59.
Williams, G.E. (1986). The Acraman impact structure: source of ejecta in late Precambrian shales, South Australia. *Science* 233, 200–203.
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Our Solar System

VC 2.0: Science Understanding — Earth and space sciences: Structure and scale of the solar system; classification of planets

5.1 The eight planets

Entry task: order the planets

Write the eight planets into the table in order from the Sun, then tick which ones have rings and which have moons. Do this **before** you look at the reference table below — use a mnemonic if it helps.

Word bank: Saturn • Mercury • Neptune • Earth • Jupiter • Venus • Uranus • Mars

From the Sun	Planet	Rings?	Moons?
1			
2			
3			
4			
5			
6			
7			
8			

What pattern do you notice about which planets have rings?

Planet	Type	Distance (AU)	Diameter (km)	Moons
[redacted]	Terrestrial	0.39	4,879	0
Venus	Terrestrial	0.72	12,104	0
Earth	Terrestrial	1.00	12,742	1
Mars	Terrestrial	1.52	6,779	2
Jupiter	Gas giant	5.20	139,820	95
Saturn	Gas giant	9.54	116,460	146
Uranus	Ice giant	19.2	50,724	28
Neptune	Ice giant	30.1	49,244	16

Mnemonic: **M**y **V**ery **E**xcited **M**other **J**ust **S**erved **U**s **N**oodles

What is an Astronomical Unit (AU)?

1 **AU** = the average Earth–Sun distance = approximately 150,000,000 km. This unit makes solar system distances much easier to write and compare. Neptune is 30 AU from the Sun — thirty times farther from the Sun than Earth.

Check your understanding *11 marks*

Q5.1 (2 marks) Using the table, name the four terrestrial planets.

Q5.2 (1 mark) Which planet has the most moons, and how many does it have?

Q5.3 (1 mark) State the type given in the table for Uranus and Neptune.

Q5.4 (2 marks) Convert Jupiter's distance from the Sun (5.20 AU) into kilometres.

Q5.5 (2 marks) Calculate how many times farther from the Sun Neptune is than Earth.

Q5.6 (3 marks) Compare the terrestrial planets with the gas giants, using two pieces of evidence from the table.

5.2 Other solar system objects

- **Asteroid belt**: between Mars and Jupiter; rocky/metallic objects; gravitational influence of Jupiter prevented these from forming a planet
- **Dwarf planets**: Pluto, Ceres, Eris, Haumea, Makemake — too small and haven't cleared their orbital neighbourhood
- **Kuiper Belt**: beyond Neptune (30–50 AU); icy objects including Pluto

- **Comets**: icy bodies; develop a tail of gas and dust when they approach the Sun
- **Oort Cloud**: distant spherical shell of icy objects; source of long-period comets

Check your understanding *9 marks*

Q5.7 (1 mark) State where the asteroid belt is located.

Q5.8 (2 marks) Explain why the objects in the asteroid belt never formed into a planet.

Q5.9 (1 mark) State the distance range of the Kuiper Belt in AU.

Q5.10 (3 marks) Name two dwarf planets and state the two reasons they are not classified as planets.

Q5.11 (2 marks) Describe what happens to a comet as it approaches the Sun.

Space Technology and Sustainability

Part A — Exploration — History and Technology

VC 2.0: Science Understanding – Science as a Human Endeavour: Space exploration technologies; Australian contributions

6.1 Key milestones in space exploration

Entry task: sequence the space milestones

Write the eight events into the timeline in chronological order, then say what each one achieved. Do this **before** you read the table below.

Word bank: Curiosity lands on Mars • ISS assembly begins • JWST launched • First humans on the Moon • Hubble Space Telescope launched • Sputnik 1 • Gagarin orbits Earth • Europa Clipper launched

Order	Event	What it achieved
1		
2		
3		
4		
5		
6		
7		
8		

Year	Event	Significance
1957	Sputnik 1 (USSR)	First artificial satellite
1961	Yuri Gagarin (USSR)	First human in orbit
1969	Apollo 11 (USA)	First humans on the Moon
1990	Hubble Space Telescope	Revolutionised optical astronomy
1998	ISS construction begins	First permanent human presence in space
2012	Curiosity rover lands on Mars	Searches for habitability
2021	JWST launched	Infrared view of the early universe
2024	Europa Clipper launched	Searching for habitability on Europa

Check your understanding *4 marks*

Q6.1 (2 marks) Using the table, state the year of Sputnik 1 and its significance.

Q6.2 (2 marks) Identify the two missions in the table launched to investigate habitability, and state which bodies they target.

6.2 Australia's contributions to space science

- **Parkes Radio Telescope** ("The Dish") — received and relayed live TV of the Apollo 11 Moon landing to 600 million viewers worldwide (1969)
- **CSIRO** — invented Wi-Fi technology from radio astronomy research; operates major Australian telescope networks
- **Australian Space Agency (ASA)** — established 2018; growing commercial space sector targeting A$12 billion by 2030
- **Square Kilometre Array (SKA)** — world's largest radio telescope, core in WA on Wajarri Country

Wajarri Yamatji Country and the SKA

The Square Kilometre Array is being built on **Wajarri Yamatji Country** in the Murchison region of Western Australia — chosen for its radio-quiet environment and extraordinarily dark skies. The Wajarri Yamatji have a formal Indigenous Land Use Agreement with the Australian government, including veto rights and community benefit provisions. The Wajarri people have maintained detailed sky knowledge of this Country for tens of thousands of years. The SKA is a model of how Western science can operate in genuine partnership with Traditional Custodians.

Your turn — Mission evaluation

Choose one of the following and evaluate it as a scientific mission. State: (a) the goal; (b) whether it was crewed or uncrewed and why; (c) one key discovery; (d) one limitation.

Options: Apollo 11, Hubble Space Telescope, Curiosity rover, JWST

Check your understanding *5 marks*

Q6.3 (2 marks) Explain the role the Parkes Radio Telescope played in the Apollo 11 mission.

Q6.4 (1 mark) State the everyday technology CSIRO developed out of its radio astronomy research.

Q6.5 (2 marks) State where the core of the Square Kilometre Array is being built, and give one reason that site was chosen.

Part B — Satellites, Debris and Sustainability

VC 2.0: Science Understanding – Science as a Human Endeavour: Sustainability of space use; Earth observation technologies

6.3 Space debris

Entry task: why orbital speed makes debris so dangerous

Kinetic energy is $E_k = \frac{1}{2}mv^2$, with mass in kilograms and speed in metres per second. Watch your units – the masses below are given in grams.

1. A 10 g bolt of debris in low Earth orbit, travelling at 7,800 m/s.

2. A 150 g cricket ball bowled at 45 m/s (about 160 km/h).

3. How many times more kinetic energy does the bolt carry than the cricket ball? What does that tell you about a collision in orbit?

Space debris includes defunct satellites, spent rocket stages, collision fragments, and lost equipment. Current estimates:

- $\approx$27,000 tracked objects $>$10 cm
- $\approx$500,000 objects 1–10 cm (too small to track, too large to ignore)
- $\approx$100 million fragments $<$1 cm

At orbital speeds ($\approx$7.8 km/s in LEO), even a 1 cm fragment has the kinetic energy of a hand grenade. The ISS has manoeuvred to avoid debris more than 30 times.

Check your understanding *4 marks*

Q6.6 (1 mark) State the approximate number of tracked debris objects larger than 10 cm.

Q6.7 (3 marks) Explain why a fragment only 1 cm across is dangerous to a spacecraft.

6.4 Kessler syndrome

Kessler syndrome (Donald Kessler, NASA, 1978): if orbital density exceeds a critical threshold, one collision generates debris that causes more collisions in a cascade — potentially making certain orbital shells permanently unusable. Evidence suggests the Iridium 33/Cosmos 2251 collision (2009) created $>$2,000 tracked fragments.

Check your understanding *5 marks*

Q6.8 (3 marks) Define Kessler syndrome.

Q6.9 (2 marks) State the evidence given in the booklet that this cascade risk is real.

6.5 Earth observation satellites

- **Weather satellites**: Himawari and GOES provide real-time storm tracking; forecasts beyond 24 h depend entirely on satellite data
- **Climate monitoring**: sea level rise, ice extent, deforestation, coral bleaching, wildfire extent
- **Precision agriculture**: multispectral imaging tracks crop health, water stress, soil condition
- **Emergency response**: floods, earthquakes, wildfires mapped in near-real-time
- **GPS**: 24+ satellites; positioning <1 m accuracy; underpins transport, shipping, aviation, and smartphones

Satellite data on Country

Indigenous ranger groups across Australia use Earth observation satellite data to manage Country. Multispectral satellite imagery tracks vegetation health, water sources, fire risk, and invasive species spread across remote land. The CSIRO's Digital Earth Australia programme provides this data freely to ranger programs. Traditional Owners often describe the satellite data as confirming what their ancestors already knew about Country — but now they can document it and advocate for its protection with evidence that government agencies accept.

Worked example: Kinetic energy of debris

A 10 g bolt of space debris travels at 7,800 m/s. Calculate its kinetic energy. Compare with a 150 g cricket ball bowled at 45 m/s.

$KE_{\text{bolt}} = \frac{1}{2} \times 0.010 \times 7800^2 = \frac{1}{2} \times 0.010 \times 60{,}840{,}000 =$ **304,200** J

$KE_{\text{cricket}} = \frac{1}{2} \times 0.150 \times 45^2 = \frac{1}{2} \times 0.150 \times 2{,}025 =$ **152** J

The space bolt has about **2,000**$\times$ more kinetic energy than the cricket ball.

Year 8 Extension: Active debris removal and orbital governance

Active debris removal (ADR) technologies in development include robotic arm capture (ESA ClearSpace-1), electromagnetic tether deorbit (JAXA ELSA-d), and laser ablation. The fundamental challenge is economic: removing debris benefits all space users, not just the company paying for removal — a classic "tragedy of the commons." Proposed governance solutions include binding UN agreements, debris levies on satellite operators, and mandatory deorbit insurance. Current international frameworks (Outer Space Treaty 1967, UN COPUOS guidelines) are non-binding and insufficient for the scale of the problem.

Check your understanding

7 marks

Q6.10 (3 marks) Name three distinct uses of Earth observation satellites given in the booklet.

Q6.11 (2 marks) Explain why weather forecasts beyond 24 hours depend on satellites.

Q6.12 (2 marks) State the accuracy of GPS positioning and name two sectors that depend on it.

Aboriginal and Torres Strait Islander Astronomy

VC 2.0: Science Understanding – Science as a Human Endeavour: Indigenous knowledge systems; dark constellations; ecological calendars

Context: The world's oldest astronomical traditions

Aboriginal and Torres Strait Islander peoples have practised systematic astronomy for at least 65,000 years — making these the oldest continuing astronomical traditions on Earth. This is not historical knowledge. It is a living, ongoing science embedded in ceremony, ecology, governance, and art.

7.1 Dark constellations

Western constellation traditions connect *stars* into figures. Aboriginal traditions also identify figures formed by the **dark dust lanes** of the Milky Way — regions of interstellar dust that block the light of stars behind them.

- The **Emu in the Sky**: head = Coalsack Nebula (near the Southern Cross); body traces dark lanes along the Milky Way. When the Emu's head is near the horizon at dusk in autumn, emus are laying eggs — verified independently by zoologists

Figure 1:

- Other dark constellations: Possum, Caterpillar, Pregnant Woman, Wedge-tailed Eagle
- The dark areas are **molecular clouds** — real, scientifically significant objects and birthplaces of new stars

Check your understanding *9 marks*

Q7.1 (2 marks) Explain how a dark constellation differs from a Western constellation.

Q7.2 (2 marks) Name the nebula that forms the head of the Emu in the Sky, and the constellation it lies near.

Q7.3 (3 marks) Explain what the dark lanes of the Milky Way actually are, and why they are scientifically significant.

Q7.4 (2 marks) Describe the ecological information encoded by the position of the Emu in the Sky in autumn.

7.2 The Boorong star catalogue

The Boorong people (Wergaia, north-western Victoria) have one of the best-documented Aboriginal astronomical knowledge systems. With the assistance of Elder William Barak, surveyor William Stanbridge documented 47 named celestial objects in 1857:

- The Magellanic Clouds (as *Yugarilya*)
- Open star clusters (Pleiades, Hyades)
- Variable stars
- A record of the Eta Carinae brightening event of c.1843 (*Collowgulloric War*) — consistent with European records of the same event

Check your understanding *9 marks*

Q7.5 (3 marks) State how many celestial objects were documented in the Boorong catalogue, in what year, and by whom.

Q7.6 (2 marks) Name two types of celestial object recorded in the Boorong catalogue.

Q7.7 (1 mark) State the Boorong name for the Magellanic Clouds.

Q7.8 (3 marks) Explain why the Boorong record of the Eta Carinae brightening is scientifically significant.

7.3 The sky as ecological calendar

Wurundjeri sky knowledge in practice

On Wurundjeri Country, the rising of the Pleiades (Seven Sisters, *Nurrumbunguttias*) marks the transition between seasons and signals specific ecological activities. The six-season Kulin calendar integrates the position and orientation of stars, the Milky Way, and the Moon to create a precise framework for land management that has been maintained across hundreds of generations.

The seasonal position of the Emu in the Sky has been validated by ecologists: when the Emu's body is horizontal above the southern horizon at dusk in autumn, emus are nesting and their eggs are available. This ecological timing is accurate and has been used for at least 65,000 years.

Your turn — Evaluating oral knowledge

A student argues: "Aboriginal astronomy can't be reliable science because it's not written down." Write a three-point rebuttal with specific evidence.

Year 8 Extension: Oral knowledge systems and scientific reliability
Research suggests Aboriginal oral traditions have preserved geographically accurate information about sea levels from 10,000–12,000 years ago (Nunn & Reid, 2016), volcanic eruptions from 6,000–8,000 years ago (Mount Eccles, Gunditjmara traditions), and astronomical transient events from the 1840s (Boorong/Eta Carinae). Multi-layered encoding (song, story, ceremony, art, kinship) creates a redundant error-correction system that may, over long timescales, be more robust than single-medium written records.

Check your understanding *8 marks*

Q7.9 (2 marks) Explain what the rising of the Pleiades signals on Wurundjeri Country.

Q7.10 (3 marks) Describe two ways the six-season Kulin calendar differs from the European four-season model.

Q7.11 (3 marks) Evaluate the claim that Aboriginal astronomy is a science. Use one specific piece of evidence from this lesson.

Above level

Extension sections

The three sections that follow sit **above** the Year 7 level. The core unit — Sections 1 to 6 — is complete without them.

Work through them if you have finished the earlier sections, or if your teacher sets them as extension.

The Search for Life — Astrobiology

Above level — extension. This section sits **above** the Year 7 level. The core unit is complete without it — work through it if you have finished the earlier sections, or if your teacher sets it as extension.

VC 2.0: Science Understanding – Biological sciences: Conditions for life; Earth and space sciences: Habitable zones; exoplanets

8.1 Requirements for life as we know it

Entry task: what does life actually need?

Sort the eight conditions into the two columns, then rank the four you think matter most. Do this **before** you read on — you are predicting, not looking it up.

Sort these: liquid water • oxygen atmosphere • carbon chemistry • sunlight • an energy source • a magnetic field • a stable environment over time • moderate temperature

Required for life as we know it	Not required (but might help)

My top four, most important first — and why:

Life on Earth requires:

- **Liquid water**: as a solvent for biochemical reactions
- **Carbon chemistry**: carbon forms 4 bonds and chains of any length — uniquely suited to the complexity life requires (CHNOPS elements: Carbon, Hydrogen, Nitrogen, Oxygen, Phosphorus, Sulfur)
- **Energy source**: sunlight (photosynthesis) or chemical energy (chemosynthesis)
- **Stable environment**: over geological timescales for evolution

Check your understanding *4 marks*

Q8.1 (2 marks) List the four requirements for life as we know it.

Q8.2 (2 marks) Explain why carbon is uniquely suited to the chemistry of life.

8.2 The habitable zone

The **habitable zone** ("Goldilocks zone") is the range of distances from a star where liquid water can exist on a planet's surface. Earth is well within the Sun's habitable zone. Mars is on the outer edge.

Beyond the habitable zone: tidal heating

Tidal heating — gravitational flexing by a parent planet — can maintain liquid water far outside the classical habitable zone. This is why Europa (Jupiter's moon) and Enceladus (Saturn's moon) are high-priority astrobiology targets despite being far from the Sun.

Check your understanding *5 marks*

Q8.3 (2 marks) Define the habitable zone and state where Earth and Mars sit relative to it.

Q8.4 (3 marks) Explain how tidal heating allows liquid water to exist far outside the classical habitable zone.

8.3 Promising targets in our solar system

- **Mars**: ancient river channels, delta deposits, and lake beds confirm liquid water billions of years ago. Possible present-day liquid brines under the southern polar ice cap. Perseverance rover collecting samples for return to Earth
- **Europa**: 10–30 km ice shell over a global saltwater ocean; tidal heating from Jupiter; probable hydrothermal vents on ocean floor. Europa Clipper (NASA, 2024) will flyby 49 times
- **Enceladus**: active water geysers from cracks in its icy shell; Cassini detected H_2, CO_2, and organic molecules in the plume — a potential chemical energy source for life

Check your understanding *8 marks*

Q8.5 (2 marks) State two pieces of evidence that Mars once had liquid water.

Q8.6 (3 marks) Describe the structure of Europa that makes it a high-priority astrobiology target.

Q8.7 (3 marks) State what Cassini detected in the plume of Enceladus, and explain why this matters for the search for life.

8.4 Extremophiles

Extremophiles are organisms that thrive in conditions once thought impossible for life:

- Hydrothermal vents (no sunlight, extreme pressure, 400°C water)
- Antarctic ice (-20°C)
- Highly acidic environments (pH 0)
- Highly radioactive environments (Deinococcus radiodurans)

Their existence vastly expands the range of environments where we search for life elsewhere.

Check your understanding *5 marks*

Q8.8 (3 marks) Define "extremophile" and give two examples of the extreme conditions they tolerate.

Q8.9 (2 marks) Explain how the discovery of extremophiles changed where scientists search for life beyond Earth.

8.5 Exoplanets and biosignatures

More than 5,500 exoplanets confirmed (2024). Detection methods:

- **Transit method**: planet passes in front of star; brightness dips slightly; detects size and orbital period
- **Radial velocity**: planet makes star wobble; detected as Doppler shifts; detects mass

Biosignatures: chemical signs of life detectable at a distance. Oxygen + methane together are a strong biosignature (both are destroyed by each other unless continuously replenished by living processes). JWST has already detected CO_2 on exoplanet WASP-39b; detecting O_2 and CH_4 together on an Earth-like planet is the next frontier.

Sky beings and the question of origins

In Wurundjeri tradition, the creator being Bunjil the Wedge-tailed Eagle flew into the sky after establishing Country and became the star Altair. His sons, the Bram-bram-bult, continue to observe from the sky. Aboriginal Australian cosmologies have always understood the sky as inhabited and relational — a different, valid framework for the question of what exists beyond our world. Western astrobiology asks whether microbial life exists in Europa's ocean. Aboriginal cosmology asks a different kind of question: about origin, relationship, and meaning. Both are genuine human responses to the same remarkable sky.

Worked example: Drake equation

Using the Drake equation with optimistic values: $R_* = 3$/yr, $f_p = 1$, $n_e = 0.1$, $f_l = 1$, $f_i = 0.1$, $f_c = 0.1$, $L = 10{,}000$ *years. Estimate N (civilisations in the Milky Way).*

$N = 3 \times 1 \times 0.1 \times 1 \times 0.1 \times 0.1 \times 10{,}000 =$ **30**

With pessimistic estimates (especially short L), N could be less than 1. The equation is a framework for identifying what we don't know rather than a precise answer.

Your turn — Life on Europa

Construct a scientific argument for whether life might exist in Europa's subsurface ocean. Include: (a) conditions present; (b) an Earth analogue; (c) missing evidence; (d) how you would obtain it.

Check your understanding

7 marks

Q8.10 (1 mark) State approximately how many exoplanets had been confirmed by 2024.

Q8.11 (3 marks) Explain the transit method of exoplanet detection and state what it reveals about the planet.

Q8.12 (3 marks) Explain why oxygen and methane detected together are considered a strong biosignature.

Year 8 Extension: The Fermi Paradox
Given the size and age of the universe, intelligent life should be detectable — yet we detect nothing. Proposed resolutions include: (1) Rare Earth hypothesis (complex life is extraordinarily rare); (2) Great Filter (a catastrophic barrier that most civilisations don't survive); (3) communication limits (light-speed travel; we've only been broadcasting for ∼100 years in a galaxy 100,000 light-years across); (4) non-radio technologies we don't recognise. The Breakthrough Listen initiative uses Parkes and other telescopes in systematic SETI searches. No confirmed signal has ever been received.

Gravity and Orbits

Above level — extension. This section sits **above** the Year 7 level. The core unit is complete without it — work through it if you have finished the earlier sections, or if your teacher sets it as extension.

VC 2.0: Science Understanding – Physical sciences: Gravitational force; orbital motion; Newton's law of universal gravitation

9.1 Newton's law of universal gravitation

Every object with mass attracts every other object with mass:

$$F = \frac{G \cdot m_1 \cdot m_2}{r^2}$$

Where: F = gravitational force (N), G = gravitational constant (6.67×10^{-11} N m^2/kg^2), m_1, m_2 = masses (kg), r = distance between centres (m).

- Double either mass $\rightarrow$ force doubles
- Double the distance $\rightarrow$ force decreases to **one quarter** (inverse square law)
- Gravity is always *attractive* — it only pulls, never pushes

Check your understanding *7 marks*

Q9.1 (3 marks) Write Newton's law of universal gravitation and state what each symbol represents.

Q9.2 (1 mark) State what happens to the gravitational force between two objects if one of the masses is quadrupled.

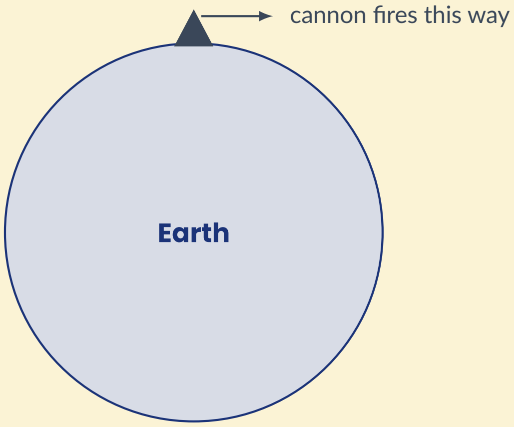
Q9.3 (2 marks) Explain what happens to the gravitational force if the distance between two objects is doubled, and name this relationship.

Q9.4 (1 mark) A student claims gravity can push objects apart. Explain why this is incorrect.

9.2 Orbital motion

Entry task: Newton's cannonball

A cannon sits on top of an impossibly tall mountain and fires horizontally. On the diagram, sketch the path of the cannonball when it is fired **(a)** slowly, **(b)** faster, and **(c)** fast enough that it never lands. Label each path.



Is the cannonball in path (c) still falling? Explain.

An **orbit** occurs when gravity provides the centripetal force needed to keep an object moving in a curved path. Orbiting objects are in constant **free fall** toward the body they orbit — but moving forward so fast they keep missing it.

Higher altitude = lower orbital speed (and longer period):

Orbit	Altitude	Speed	Period
LEO (ISS)	200–2,000 km	≈7.9 km/s	≈90 min
MEO (GPS)	2,000–35,786 km	≈3.9 km/s	≈12 h
GEO (comms, weather)	35,786 km	≈3.1 km/s	24 h

Why astronauts appear weightless

The ISS orbits at ≈408 km altitude, where gravity is about 90% of surface strength. Astronauts appear weightless because they *and* the station are falling together at the same rate. "Weightlessness" is free fall — not absence of gravity.

Check your understanding

9 marks

Q9.5 (2 marks) Define an orbit, referring to gravity and centripetal force.

Q9.6 (3 marks) Explain why an object in orbit is described as being in constant free fall.

Q9.7 (2 marks) State the approximate altitude of the ISS and the strength of gravity there as a percentage of its strength at Earth's surface.

Q9.8 (2 marks) A student says astronauts float on the ISS because there is no gravity in space. Correct this statement.

Worked example: Weight at ISS altitude

An astronaut weighs 700 N on Earth (radius 6,371 km). What is their weight at ISS altitude (408 km above surface)?

$r_{\text{ISS}} = 6{,}371 + 408 = 6{,}779$ km

Force ratio $= \left(\frac{6{,}371}{6{,}779}\right)^2 = 0.883$

Weight at ISS $= 700 \times 0.883 =$ **618** N (about 88% of surface weight)

Year 8 Extension: Escape velocity

To escape a planet's gravitational field entirely: $v_e = \sqrt{\frac{2GM}{r}}$

Earth: 11.2 km/s Moon: 2.4 km/s Mars: 5.0 km/s Jupiter: 59.5 km/s

The Moon's low escape velocity explains why it has no atmosphere: gas molecules moving at thermal speeds can escape to space over time.

Check your understanding *8 marks*

Q9.9 (2 marks) Using the table, state the orbital speed and period of the ISS.

Q9.10 (2 marks) Describe the relationship between orbital altitude and orbital speed shown in the table.

Q9.11 (2 marks) Explain why a satellite in geostationary orbit appears to stay in the same place in the sky.

Q9.12 (2 marks) GPS satellites operate in MEO. Using the table, state the altitude range and orbital period of this orbit.

Stars, Galaxies and Starlight

Above level — extension. This section sits **above** the Year 7 level. The core unit is complete without it — work through it if you have finished the earlier sections, or if your teacher sets it as extension.

Part A — Stars and Galaxies

VC 2.0: Science Understanding – Earth and space sciences: Properties of stars; galaxies; scale of the universe

10.1 Light-years and stellar distances

Entry task: which unit fits which distance?

Complete the table. For each distance, choose the unit that gives a sensible number — kilometres, astronomical units (AU) or light-years — and fill in the blanks.

Distance	Best unit	Approximate value
Earth to the Moon		384,400 km
Earth to the Sun	AU	
To the nearest star (Proxima Centauri)		4.2 light-years
Diameter of the Milky Way	light-years	

Why would it be silly to give the Milky Way's diameter in kilometres?

A **light-year** is the distance light travels in one year: approximately 9.46×10^{12} km (9.46 trillion km). It is a unit of *distance*, not time.

- The Sun: light takes 8 min 20 s to reach Earth
- Proxima Centauri (nearest star): 4.2 light-years
- Milky Way diameter: ≈100,000 light-years
- Andromeda Galaxy: 2.5 million light-years

Looking back in time

When we observe a star 100 light-years away, we see it as it was 100 years ago — the light we detect left it then. Telescopes are time machines. JWST images galaxies as they appeared just 300–400 million years after the Big Bang.

Check your understanding *4 marks*

Q10.1 (2 marks) Define a light-year and state what type of quantity it measures.

Q10.2 (2 marks) Proxima Centauri is 4.2 light-years away. Explain what we are actually seeing when we look at it tonight.

10.2 Star types and the OBAFGKM sequence

Stars are classified by surface temperature and colour:

Class	Colour	Temperature	Example
O	Blue-violet	$>$30,000 K	Zeta Puppis
B	Blue-white	10,000–30,000 K	Rigel
A	White	7,500–10,000 K	Sirius
F	Yellow-white	6,000–7,500 K	Procyon
G	Yellow	5,200–6,000 K	**The Sun**
K	Orange	3,700–5,200 K	Arcturus
M	Red	$<$3,700 K	Proxima Centauri

Check your understanding *5 marks*

Q10.3 (2 marks) Using the table, state the spectral class and colour of the Sun.

Q10.4 (3 marks) Arrange Rigel, the Sun and Proxima Centauri in order of decreasing surface temperature, justifying your order with data from the table.

10.3 Stellar life cycle (sun-like star)

Nebula → Protostar → **Main seq.** → Red giant → White dwarf

Massive stars ($>8\ M_\odot$): Main sequence → Red supergiant → Supernova → Neutron star / Black hole

Boorong knowledge: The Magellanic Clouds

The Large and Small Magellanic Clouds — dwarf galaxies visible to the naked eye from the Southern Hemisphere — are incorporated into Boorong star lore as *Yugarilya*, the two Bram-bram-bult (sons of Bunjil the creator), who appear as wedge-tailed eagles in the sky. Aboriginal Australians observed these objects for tens of thousands of years before European astronomers named them after Magellan's 1521 voyage.

Check your understanding *4 marks*

Q10.5 (2 marks) List, in order, the stages in the life cycle of a Sun-like star.

Q10.6 (2 marks) State how the life cycle of a star more massive than 8 solar masses differs after the main sequence.

Year 8 Extension: The Hertzsprung–Russell Diagram
The H–R diagram plots stellar luminosity (vertical axis) against surface temperature (horizontal axis, running *right to left* — hot on left). Stars cluster in recognisable zones: the **main sequence** (a diagonal band), the **giant branch** (upper right, cool but luminous), and **white dwarfs** (lower left, hot but dim). A star's position reveals its evolutionary stage, mass, and age.

Part B — Light from Space — Telescopes and the EM Spectrum

VC 2.0: Science Understanding — Physical sciences: Electromagnetic radiation; wave properties; telescopes as scientific instruments

10.4 The electromagnetic spectrum

Entry task: order the electromagnetic spectrum

Write the seven regions into the strip in order, longest wavelength on the left. Then circle the one we can see with our eyes.

Word bank: gamma • infrared • microwave • radio • ultraviolet • visible • X-ray

⟵ longer wavelength, lower energy higher energy, shorter wavelength ⟶

Visible light is only a small part of the **electromagnetic spectrum** (EM spectrum). All EM waves travel at the speed of light (3×10^8 m/s) but differ in wavelength and frequency.

Region	Wavelength	Energy	Astronomical use
Radio waves	mm to km	Lowest	Pulsars, CMB, galaxy mapping (Parkes, SKA)
Microwaves	mm	Low	CMB, molecular clouds
Infrared	1 μm–1 mm	Medium-low	Star formation, cool objects, exoplanet atmospheres (JWST)
Visible	400–700 nm	Medium	Stars, planets, galaxies (optical telescopes)
UV	10–400 nm	Medium-high	Hot stars, quasars
X-ray	0.01–10 nm	High	Black holes, neutron stars, galaxy clusters
Gamma ray	$<$0.01 nm	Highest	Supernovae, gamma-ray bursts

Check your understanding *4 marks*

Q10.7 (2 marks) State one property all electromagnetic waves share, and two properties that differ between them.

Q10.8 (2 marks) Using the table, name the region of the spectrum used to study black holes, and the region used to map galaxies with the SKA.

10.5 Telescopes

- **Refracting telescopes**: glass lenses bend light to a focus. Limited to small instruments
- **Reflecting telescopes**: mirrors reflect light to a focus. All major modern telescopes; can be built much larger (Hubble: 2.4 m mirror; JWST: 6.5 m)
- **Radio telescopes**: large dish antennas; collect radio waves; Parkes dish (64 m) relayed Moon landing footage
- **Space telescopes**: above the atmosphere; avoid absorption and blurring; JWST orbits at the Sun–Earth L2 point

Check your understanding *4 marks*

Q10.9 (2 marks) Explain the difference between a refracting and a reflecting telescope.

Q10.10 (2 marks) Give two reasons for placing a telescope in space rather than on the ground.

10.6 Spectroscopy — reading starlight

When starlight passes through a prism or diffraction grating, **absorption lines** appear — dark lines at specific wavelengths absorbed by elements in the star's outer atmosphere. Each element has a unique spectral fingerprint. From absorption spectra, astronomers determine:

- Elemental **composition** of stars and exoplanet atmospheres
- **Temperature** (line types reveal temperature class)
- **Velocity**: blueshifted = moving toward us; redshifted = moving away (Doppler effect)

Wajarri knowledge and the MWA

The Murchison Widefield Array (MWA) — precursor to the SKA — sits on Wajarri Yamatji Country in Western Australia. It maps hydrogen across the Milky Way using radio waves invisible to the naked eye. The Wajarri have observed and named features of this same sky for tens of thousands of years using different instruments (their senses, memory, and song). Both the MWA and Wajarri sky knowledge reveal the same cosmos at different scales and with different tools.

Year 8 Extension: Interferometry and the Event Horizon Telescope
By linking multiple radio telescopes worldwide and combining their signals with atomic-clock precision, astronomers create a **virtual telescope** as large as the separation between dishes — a technique called **interferometry**. The **Event Horizon Telescope** linked 8 observatories to create an Earth-sized virtual dish, imaging the shadow of a black hole in galaxy M87 (2019) and the Milky Way's central black hole Sgr A* (2022). Resolution achieved: 40 microarcseconds — equivalent to reading a newspaper in New York from Sydney.

Check your understanding *4 marks*

Q10.11 (3 marks) Explain how absorption lines allow astronomers to identify the elements present in a star.

Q10.12 (1 mark) A star's spectral lines are redshifted. State what this tells you about its motion.

Unit Glossary

Absorption lines — dark lines in a stellar spectrum caused by elements absorbing specific wavelengths of light.

Astronomical unit (AU) — average Earth–Sun distance; ≈150,000,000 km.

Axial tilt — Earth's 23.5° tilt relative to its orbital plane; causes seasons.

Biosignature — chemical or physical evidence of life detectable at a distance.

Black hole — region of space where gravity is so extreme that not even light can escape.

Dark constellation — figure formed by dark dust lanes of the Milky Way, used in Aboriginal Australian astronomy.

Drake equation — formula estimating the number of communicable civilisations in the Milky Way.

Electromagnetic spectrum (EM) — the full range of electromagnetic radiation from radio waves to gamma rays.

Exoplanet — a planet orbiting a star other than the Sun.

Extremophile — organism that thrives in extreme conditions (temperature, pressure, pH, radiation).

Free fall — falling under gravity alone with no other forces; the state of orbiting objects.

Geostationary orbit (GEO) — orbit at 35,786 km; orbital period equals Earth's rotation; satellite appears stationary.

Habitable zone — range of distances from a star where liquid water can exist on a planet's surface.

Interferometry — technique combining signals from multiple radio telescopes to create a virtual large telescope.

Inverse square law — relationship where a quantity decreases with the square of the distance (gravity, light intensity).

Kessler syndrome — cascade of collisions in orbit where debris generates more debris.

Light-year — distance light travels in one year ($\approx 9.46 \times 10^{12}$ km); a unit of *distance*.

Lunar eclipse — Earth's shadow falls on the Full Moon.

Main sequence — phase of a star's life during which it fuses hydrogen in its core.

Molecular cloud — dense region of interstellar gas and dust; birthplace of new stars; forms dark constellations.

Nebula — cloud of gas and dust in space; some are remnants of dead stars, others are star-forming regions.

Nuclear fusion — process in which hydrogen nuclei combine to form helium, releasing energy; powers the Sun.

Orbit — curved path of an object around a more massive body due to gravitational attraction.

Phases (lunar) — apparent changes in the Moon's shape as seen from Earth; caused by changing view of its lit half.

Redshift / Blueshift — change in wavelength of light due to relative motion (Doppler effect).

Solar eclipse — Moon passes between Earth and Sun; Moon's shadow falls on Earth.

Solar system — the Sun and all objects gravitationally bound to it, including 8 planets, dwarf planets, moons, asteroids, and comets.

Spectroscopy — analysis of light spectra to determine composition, temperature, and velocity of astronomical objects.

Spring tides — exceptionally large tidal range when Sun, Earth, and Moon are aligned.

Supernova — catastrophic explosion of a massive star at the end of its life.

Tidal heating — heating of a moon due to gravitational flexing by a parent planet; maintains liquid water on Europa and Enceladus.

Transit method — detecting exoplanets by measuring the slight dimming of a star as a planet passes in front.

White dwarf — stellar remnant left after a sun-like star sheds its outer layers; hot but dim.

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